MIL-P-3756D
24 October 1980
SUPERSEDING
MIL-P-3756C
16 April 1975

MILITARY SPECIFICATION

PROTRACTOR, ONE-ARM: SEMICIRCULAR

(CORROSION-RESISTANT ALLOY, WITH CASE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a semicircular, metal protractor with a movable arm.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-391	- Plastic Sheets, Rods and Tubing, Rigid Cast, Methacrylate (Multiapplication).
L-P-1035	- Plastic Molding Material, Vinyl Chloride Polymer and Vinyl Chloride-Vinyl Acetate Copolymer, Rigid.
QQ-C-320	- Chromium Plating (Electrodeposited).
QQ-C-585	- Copper-Nickel-Zinc Alloy Plate, Sheet, Strip, and Bar (Copper Alloy Numbers 735, 745, 752, 762, 766 and 770).

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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QQ-C-586	- Copper-Nickel-Zinc Alloy; Rod, Shapes and Flat Products with Finished Edges (Flat Wire, Strip, and Bar).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-5425	- Plastic, Sheet Acrylic, Heat Resistant.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The protractor shall be as shown in Figure 1 and as specified herein (see 6.3).

3.2 Material. The protractor, including the arm and clamp, shall be either nickel-silver conforming to QQ-C-586, half-hard, Copper Alloy No. 752; stainless steel conforming to ASTM A 167 Type 302; or nickel-brass conforming to QQ-C-585, half-hard, Copper Alloy No. 766 (see 6.4).

3.3 Protractor. The protractor shall consist of a semicircular arc 8 inches in diameter, graduated in 1/2-degree intervals with a crossbar, a centerpiece, a movable arm, clamp, and tangent screws, and shall be provided with a case. The inner edge of the crossbar shall be not less than 3/32 inch below the horizontal centerline of the protractor (see Figure 1).

3.3.1 Graduations and numbering. The graduations and numbering shall be cut into the surface of the protractor on the outer circular edge, and shall be machine divided. The length of graduations shall be apportioned as shown on Figure 1. Graduation lines shall be 0.005 inch plus or minus 0.001 inch wide at plus 72° F, and all graduation lines shall be within 0.0012 inch of their true relative position. All graduation marks shall be V-shaped, uniform, and 0.005 inch deep. The arc of graduation lines shall extend at least 15 degrees below the horizontal centerline.

3.3.2 Centerpiece. The centerpiece of the protractor shall be plastic, not less than 0.060 inch thick and 1/2 inch in diameter, conforming to MIL-P-5425, Finish A, or L-P-391, Item A, Type I, Grade A. The center of the protractor shall be indicated by the intersection of two straight lines perpendicular to each other engraved on the lower surface of the centerpiece. The projected horizontal line shall intersect the 0- and 180-degree marks on the protractor, and the projected vertical line shall intersect the 90-degree mark. To reduce parallax, the lower surface of the centerpiece with the engraved lines shall be not more than 0.015 inch above the table in normal use.

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3.4 Arm. The arm shall pivot about the center of the protractor and shall extend at least 6 inches beyond its outer edge. The arm shall be provided with a double vernier graduated to read directly to 1 minute of arc. Width and tolerance of the graduation marks on the vernier shall exactly match the graduation marks on the protractor. A small knob shall be placed near the tip of the arm to provide a convenient means of rotating the arm.

3.4.1 Clamp and tangent screw. A clamp and tangent screw shall be provided. The clamp shall be activated by a thumbscrew and shall hold the arm in any set position throughout the working range of the protractor. When released, the clamp shall slide freely with the arm. The tangent screw shall provide a positive and precise means of setting the vernier. The clamp lock and tangent screws shall be the same material as the protractor.

3.5 Finish. All metal, and plastic surfaces, except screw threads and knurled edges, shall have a smooth satin finish. Graduations, numbers, and marking shall be filled with permanent black paint or ink.

3.6 Case. The protractor shall be provided with a fitted plastic case, with hinged cover, conforming to L-P-1035, Composition A, Type III, Grade GU, color black. The case shall be lined with a soft fungus-resistant material to protect the instrument during storage and transit.

3.6.1 Fasteners and hinges. Fasteners and hinges shall be corrosion-resistant metal, or shall be chromium plated in accordance with QQ-C-320, Class 1, Type II.

3.7 Identification marking. Each protractor shall be identified in accordance with MIL-STD-130.

3.8 Workmanship. All parts of the protractor shall be clean and free from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Each lot shall be identified.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.1. AQL shall be 2.5 percent defective.

4.4 Inspection procedures.

4.4.1 Examination. The protractor shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Components missing or not as specified.
103. Shape, size or dimensions not as specified.
104. Finish not as specified.
105. Workmanship not as specified.

4.4.2 Test.

4.4.2.1 Graduations. The graduations on the protractor shall be checked against a series of angles laid out on a standard graduation metal or plastic plate controlled by a Government-approved calibration system. Any deviation greater than 0.0012 inch shall constitute failure of this test.

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4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106.	Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered a defect.
107.	Consolidation not as specified for Level A.
108.	Gross weight of wood boxes exceeds 200 pounds for Level A.
109.	Gross weight or size of fiberboard boxes exceeds the limitations for Level B.
110.	Fiberboard boxes not reinforced with tape as specified for Level B.
111.	Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each protractor contained in its case shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.1.1 Consolidation. Protractors preserved as specified in 5.1.1 shall be consolidated together, in quantities not to exceed 25, in a box conforming to PPP-B-636, W5c.

5.1.2 Commercial. Each protractor contained in its case shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

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5.2.1 Level A. The protractors, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional or PPP-B-621, Class 2, style optional. The gross weight of each box shall not exceed 200 pounds. Boxes shall be closed and strapped in accordance with the appendix to the applicable box specification.

5.2.2 Level B. The protractors, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-636, V3c or V3s, style optional. The gross weight or size of each box shall not exceed the limitations of the box specification and the boxes shall be closed and reinforced with bands of reinforced tape in accordance with the appendix to the box specification.

5.2.3 Commercial. The protractors, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (Level A and Level B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The protractor is used for measuring and laying out angles in topographic mapping and general drafting room practice.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Information figure. Figure 1 shows a type of protractor which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of other designs which conform to this specification.

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6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

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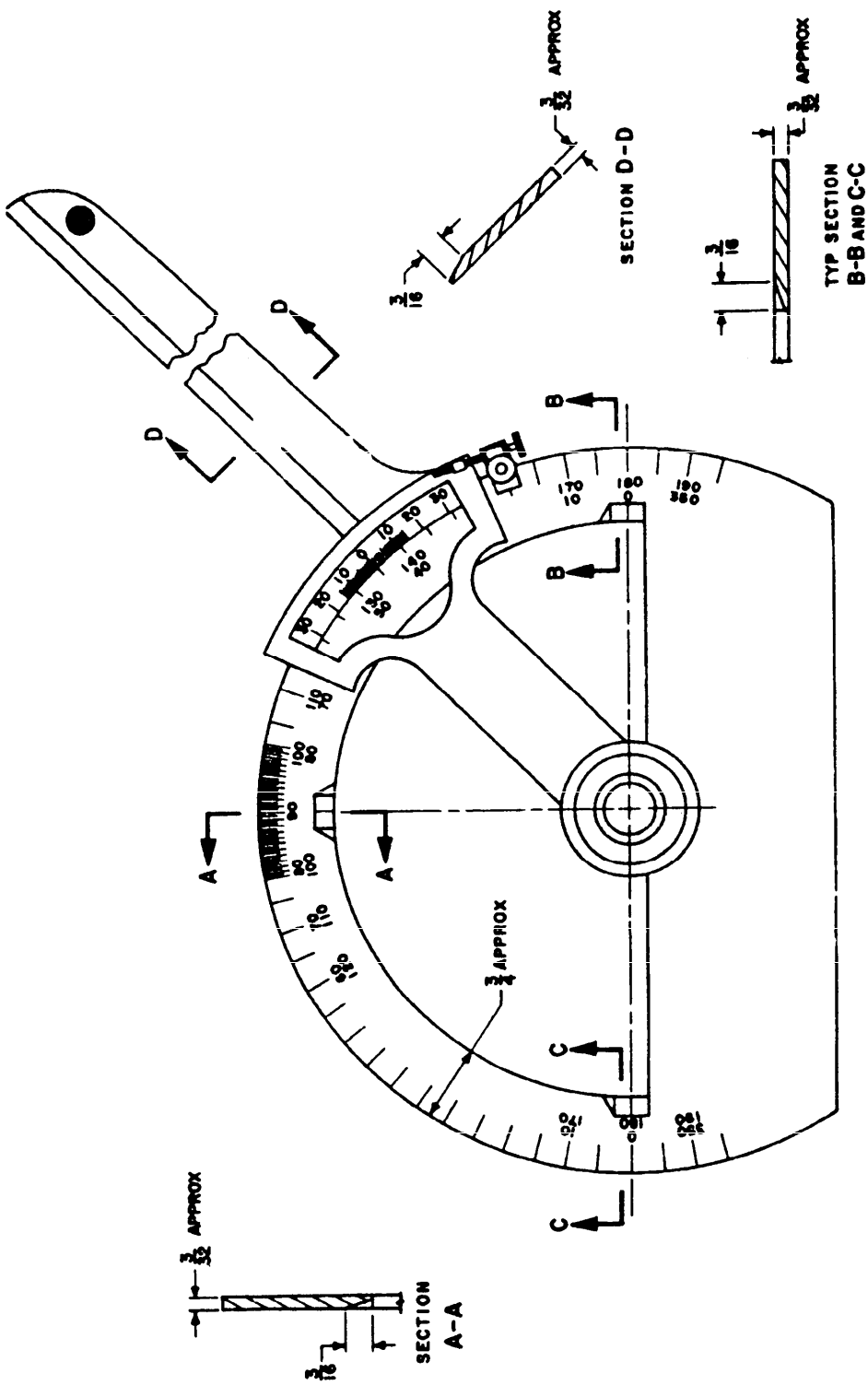


FIGURE 1. Protractor, semicircular, corrosion resistant metal.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-P-3756D Protractor, One-Arm: Semicircular (Corrosion-Resistant Alloy, with Case)

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